

claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-a80d907a5a78f688a.jsonl`  
- SHA-256: `ae29a94b6822c47a042bf0754f71fd4725ff57ca53dc28c2bc23fbb52de3e012`  
- Source modified: `2026-06-06T17:38:49+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_6237b110-d58`  
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:37:12.153Z)

Adversarial Claim Verifier (voter 3/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):

- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:

1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ?

especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?

5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio).

Claim under review

"Audio-based ball-hit detection is unreliable in isolation: their fully-automatic audio segmentation (speech/applause/ball-hits/noise/music classifier) achieved only 76.9% recall and 46.3% precision, with noise frequently misclassified as ball hits ? and these extraction errors propagated downstream, dragging the audio-only structuring below the visual-only system."

****Source:**** <https://www.irisa.fr/metiss/gravier/biblio/03/kijak-icme03.pdf> (primary)

****Supporting quote:**** "The approach using only audio features automatically segmented gives poor performance. Recall rate for audio segmentation as described in 2.3 is 76.9%, and precision is 46.3%. Ball hits and music are well classified, whereas noise is often classified as ball hits. The errors from the automatic audio segmentation spread over the structuration process and drag the performance down... An improvement was obtained by estimating the HMM parameters using automatically generated audio features, but it did not outclass the video only system."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-06T17:37:19.077Z)

I'll verify this claim about audio-based ball-hit detection from the Kijak ICME03 paper. Let me start by fetching the primary source.

3. user (2026-06-06T17:37:29.217Z)

I appreciate your detailed questions, but I must inform you that the content provided appears to be a corrupted or improperly encoded PDF file. The text consists largely of binary/compressed data that cannot be reliably parsed for the specific information you're requesting.

To accurately extract:

1. Exact recall and precision numbers for audio segmentation
2. Audio classifier classes (speech, applause, ball hits, noise, music)
3. Noise misclassification patterns
4. Audio-only vs visual-only system conclusions
5. Sport identification (tennis vs other)
6. Publication year and methods (HMM)

I would need a properly formatted, readable version of the document. The current file format prevents me from locating and quoting the relevant passages you've requested.

****Recommendation:**** Please provide either:

- A text-extracted version of the PDF
- A properly decoded version of the document
- The original source document in a readable format

Once you supply a readable version, I can extract the specific performance metrics, methodology

details, and conclusions with the exact quotations you've requested.

[Binary content (application/pdf, 188.7KB) also saved to C:\Users\avidu\.claude\projects\C-Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\tool-results\webfetch-1780767080102-6svilu.pdf]

4. assistant (2026-06-06T17:37:31.966Z)

The PDF was downloaded but couldn't be parsed by WebFetch. Let me read it directly since it's saved locally.

5. user (2026-06-06T17:37:32.736Z)

```
pdftoppm failed: Command 'pdftoppm' not found or is in an unsafe location (current directory)
```

6. assistant (2026-06-06T17:37:35.225Z)

The PDF reader needs pdftoppm. Let me try to extract text from the PDF using available tools.

7. user (2026-06-06T17:37:36.696Z)

pypdf available

8. user (2026-06-06T17:37:41.010Z)

Traceback (most recent call last):

```
File "<string>", line 3, in <module>
    r = pypdf.PdfReader('/mnt/c/Users/avidu/.claude/projects/C--Users-avidu-Projects-badminton-
highlight-indexer/42211f80-2169-4025-ad9d-99c1aa29ff70/tool-
results/webfetch-1780767080102-6svilu.pdf')
```

```
File "C:\Users\avidu\AppData\Local\Programs\Python\Python313\Lib\site-
packages\pypdf\_reader.py", line 151, in __init__
    self._initialize_stream(stream)
```

```

~~~~~AAAAAAAAAAAA
File "C:\Users\avidu\AppData\Local\Programs\Python\Python313\Lib\site-
packages\pypdf\_reader.py", line 172, in _initialize_stream
    with open(stream, "rb") as fh:
~~~~~AAAAAAAAAAAA

```

```
FileNotFoundError: [Errno 2] No such file or directory: '/mnt/c/Users/avidu/.claude/projects/C--Users-avidu-Projects-badminton-highlight-indexer/42211f80-2169-4025-ad9d-99c1aa29ff70/tool-results/webfetch-1780767080102-6svilu.pdf'
```

9. assistant (2026-06-06T17:37:43.860Z)

The Bash tool here is using Windows Python, not WSL. Let me use the Windows path.

10. user (2026-06-06T17:37:45.998Z)

=== PAGE 1 ===

HMM BASED STRUCTURING OF TENNIS VIDEOS USING VISUAL AND AUDIO CUES

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```

ABSTRACT

This paper focuses on the use of Hidden Markov Models (HMMs) for structure analysis of videos, and demonstrates how they can be efficiently applied to merge audio and visual cues. Our approach is validated in the particular domain of tennis videos. The basic temporal unit is the video shot. Visual features are used to characterize the type of shot view. Audio features describe the audio events within a video shot. The video structure parsing relies on the anal-

ysis of the temporal interleaving of video shots, with respect to prior information about tennis content and editing rules. As a result, typical tennis scenes are identified. In addition, each shot is assigned to a level in the hierarchy described in terms of point, game and set.

1. INTRODUCTION

Video structure parsing consists in extracting logical story units from the considered video. It's a mandatory step to efficiently organize and retrieve video contents. The structure to be estimated relies on the nature of the video. In this paper, we address the problem of recovering sport video structure that can be useful in a further indexing task. Sport video analysis is motivated by the broadcasters needs of a detailed annotation of video contents. This annotation is usually used to select relevant excerpts in order to easily build summaries or magazines. Up to now, this logging task is performed manually.

Broadcasted video data consist of several multimodal information streams. Numerous approaches presented in the literature have discussed how to extract high-level semantic events from video. To achieve such a goal, algorithms have to be dedicated to one particular type of videos such as sports or news. In this context, most of the approaches use the individual visual or audio cues. Only a few attempts dealing with multimodal analysis of audio-visual information have appeared recently, such as [1] on sports, or [2] on news. These approaches rely on a successive use of visual and audio features. In a first step, visual features are used to detect plays in the original flow. In a second step, a detection of audience or commentator excitement is used to select the most exciting plays [3]. The reverse scheme has also been proposed in [4]: cheering sounds are first extracted to locate an important event. Visual analysis is then performed in cheering sounds regions to recognize the event.

In this paper, we propose a tennis scene classification and segmentation method that automatically labels each segment of a tennis video with one of the following four types: first missed serve, rally, replay, and break. Hidden Markov Models (HMMs) are used to merge audio-visual informations, and to represent the hierarchical structure of a tennis match. Domain knowledge is implicitly taken into account through the learning process associated with HMMs. This implicit approach is more suitable than heuristic rules oriented schemes because the learning process can derive more complete knowledge. In addition, previous works that use HMMs for video parsing or segmentation suggest good potential, even if only visual features are considered in sport domain [5, 6].

Our solution differs from existing methods in the following main points: (1) it combines visual/audio cues simultaneously; (2) different levels of classifications are generated over the entire tennis video at different levels; and (3), the segmented content are semantically meaningful to users.

This paper is organized as follows. Section 2 gives an overview of the system and briefly describes the exploited audio and visual features. Section 3 introduces the structure analysis mechanism. Experimental results are presented and discussed in section 4. Section 5 provides the concluding remarks.

2. SYSTEM OVERVIEW

In this section, we briefly describe the extraction of audio and visual cues. First, the video stream is automatically segmented into shots by detecting cuts and dissolve transi-

tions. For each shot, we extract from the image stream, one

=== PAGE 2 ===

Video program

frames

Shot Detection

Shot Features

Extraction

HMM Training

HMM Parsing

Observations

Program Syntax

States

Transitions

Domain Knowledge

Sport game semantics

&

Sport video syntax

Fig. 1. Structure analysis system overview

keyframe along with its image features (color and motion intensity), and from the audio stream, sound classes such as ball hits, applause, or speech. The segmented video results in an observation sequence which is parsed by a HMM process (Figure 1). As a result, each shot is assigned to a level in the hierarchy described in terms of point, game and set, and is labelled with one of these four types: first missed serve, rally, replay, and break.

Such a tennis scene classification and segmentation is possible because tennis video is characterized by a typical production style, known as the tennis video syntax.

2.1. Tennis Syntax

Sport video production is characterized by the use of a limited number of cameras at almost fixed positions. Considering a given instant, the point of view giving the most relevant information is selected by the producer, and broadcast. Therefore sports are composed of a restricted number of typical scenes producing a repetitive pattern. For example, during a rally in a tennis video, the content provided by the camera filming the whole court is selected (we name it global view in the following). After the end of the rally, the player who has just carried out an action of interest is captured with a close-up. As close-up views never appear during a rally but right after or before it, global views are generally significant of a rally. Another example consists in replays that are notified to the viewers by inserting special transitions.

We identify here four typical pattern in tennis videos that are: first missed serve, rally, replay, and break. The break class includes the scenes unrelated to games, such as commercials. A break is characterized by an important succession of such scenes. It appears when players change ends, generally every two games. We also take advantage

of the well-defined and strong structure of tennis broadcast. On the contrary of time-constrained sports that have a relatively loose structure, tennis is a score-constrained sport that can be broken down into sets, games and points (Figure 2). The different types of views present in a tennis video can be divided into four principal classes: global, medium, Fig. 2. Structure of Tennis Game

close-up, and audience. In a tennis video production, global views contain much of the pertinent information. The remaining information relies on the presence or the absence of non global views but is independent of the type of these views.

2.2. Visual Features

We used visual features to identify the global views within all the extracted keyframes. The process can be divided into two steps. First, a keyframe K_{ref} representative of a global view is selected without making any assumption about the playing area color. Once K_{ref} has been found, each keyframe is characterized by a similarity distance to K_{ref} . In the following, we motivate the choice of the visual features we use, before briefly describing the identification process. Global views are characterized by a rather homogeneous color content (the colors of the court and its surrounding), although medium and audience views are characterized by scattered color content. In addition, a global view must capture at each time the main part of the court, whereas in close-up views, the camera is generally tracking the player. The first class can thus be characterized by a small camera motion while the second implies important camera translations. Based on these observations, we choose two features to identify global views: (1) a vector of dominant colors F and its spatial coherency C , and (2) activity A that reflects camera motion during a shot.

The visual similarity measure $v(K_1, K_2)$ between two keyframes K_1 and K_2 is defined as a weighted function of the spatial coherency, the distance function between the dominant color vectors, and the activity:

$$v(K_1, K_2) = (1)$$

$$w_1 |C_1 - C_2| + w_2 d(F_1, F_2) + w_3 |A_1 - A_2|$$

where w_1 , w_2 , and w_3 are the weights.

=== PAGE 3 ===

To find a keyframe K_{ref} representative of a global view, we consider dominant colors ratios. In a reduced subset of keyframes whose highest percentage of dominant color is more than 50%, we apply the least median square method, to minimize the median distance of all the remaining keyframes to a randomly selected keyframe. This leads to obtain K_{ref} (see details in [7]). The visual similarity measure $v(K_t, K_{ref})$, denoted by v_t in the following, is then computed between each keyframe K_t and K_{ref} .

2.3. Audio Features

Generally speaking, there are two approaches to combine audio and visual features for structure analysis: one is to combine audio and visual features into a single audiovisual feature vector before the classification. The other consists in classifying separately according to each modality before integrating the classification results.

In this work, an intermediate strategy is used which consists in extracting separately high level audio and visual cues, the classification being made using the audio and visual cues simultaneously. As mentioned previously, the video stream is segmented in a sequence of shots. Since shot boundaries are more suitable for structure analysis than boundaries extracted from the soundtrack, the shot is considered as the base entity and features describing the audio content

for each shot are used to provide additional information. For each shot, a binary vector at describing which audio classes, among speech, applause, ball hits, noise and music, are present in the shot is extracted from an automatic segmentation of the audio stream. The automatic soundtrack segmentation is carried out using a Viterbi based system with Gaussian mixtures for each sound class and combination of sound classes. The soundtrack is segmented independently and the vectors at are created from this independent segmentation.

3. STRUCTURE ANALYSIS

Prior information is integrated by deriving syntactical basic elements from the tennis video syntax. We define four basic structural units: two of them are related to game phases (first missed serves and rallies), the two others denote non-game phases (breaks and replays). Each of these units is modelled by a HMM. These HMMs rely on the temporal relationships between shots.

Each HMM state models either a single shot or a dissolve transition between shots. The observation consists of three elements: the visual similarity vt between the shot keyframe and $Kref$, the shot duration dt , and the audio vector at which characterizes the presence or absence of the pre-determined audio events. More formally, the probability of an observation ot conditionally to state j is then given by $b_j(ot) = p(vt|j) p(dt|j) P[at|j]$ (2)

MATCH

S
Br
S
Br
S

G

Br
G
G
Br
G
G
Br
G
G

P

P
P
P
P

SET

BREAK

First

serve

and rally

Replay

Rally

GAME

Fig. 3. Content hierarchy of broadcast tennis video where $p(v_t|j)$ and $p(dt|j)$ are given by a smoothed histogram, and $P[at|j]$ is the product over each sound class k of the discrete probability $P[at(k)|j]$. Segmentation and classification of the whole observed sequence $o = o_1o_2...o_T$ into the different structural elements are performed simultaneously using a Viterbi algorithm. The most likely state sequence $s = s_1...s_T$ is given by

$$s = \arg \max_s \ln p(s) + \sum_t \ln bst(o_t) \quad (3)$$

To take into account the long-term structure of a tennis game, the four HMMs are connected to a higher-level HMM which represents the tennis syntax as illustrated in Figure 3. The transition probabilities between states of the higher-level HMM result entirely from a priori information, while the transition probabilities of the sub-HMMs are estimated.

4. EXPERIMENTAL RESULTS

Experimental data are composed of about 2 hours of manually labelled tennis video. About half of the data is used to train the HMM while the remaining half is reserved for the tests.

Recall and precision rates are given in Table 1 for three experiments: visual features only, audio features only and combined audiovisual approach. Shot duration is considered for all experiments. Experiments using audio features are sub-divided into two parts: using audio features manually annotated (denote by γ_{man} in Table 1) to validate the model, and using audio features extracting from automatic segmentation (denote by γ_{segm} in Table 1). The audio probabilities are estimated from the manually annotated audio features.

=== PAGE 4 ===

Visual Audio features Audio-visual
features man. segm. man. segm.
Segmentation 71% 68% 44% 78% 66%
accuracy

Classification precision recall precision recall precision recall precision recall precision recall
recall

First serve 67% 63% 92% 66% 66% 5% 87% 75% 71% 41%
Rallies 97% 54% 95% 63% 62% 28% 96% 83% 82% 65%
Replay 100% 80% 68% 38% 75% 15% 81% 88% 86% 82%
Break 51% 93% 91% 81% 30% 90% 94% 81% 68% 92%

Table 1. Classification and segmentation results with visual features only, audio features only and both audio-visual features

We observed that one single stream cannot provide enough information to identify different scenes. Rallies high precision rate (97%) and breaks recall rate (93%) show that low level visual features are efficient to identify global views from others. However, these features are not sufficient to distinguish a rally from a simple view of the court or from a first serve, as indicate by rallies low recall rate (54%). This means also that many global views are missed using only visual features. Replays detection relies essentially on dissolve transitions identification. The high precision rate corresponds to correctly detected dissolve transitions. Classification using reliable audio features increases precision rates for first missed serve. Comparable results are

obtained for rallies detection. This suggests that audio features are effective to describe rally scenes. Indeed, a rally is essentially characterized by the presence of ball hits and applause which happen at the end of the exchange, although a missed 1st serve is only characterized by the presence of ball hits. Therefore, there are less confusion between missed 1st serves and rallies. Breaks precision rate increases also. Since break is the only state characterized by the presence of music, the break state works in this case like a commercial detector, and avoids false detections. On the contrary, replays are not characterized by a representative audio content, and almost all replays are missed.

The approach using only audio features automatically segmented gives poor performance. Recall rate for audio segmentation as described in 2.3 is 76.9%, and precision is 46.3%. Ball hits and music are well classified, whereas noise is often classified as ball hits. The errors from the automatic audio segmentation spread over the structuration process and drag the performance down. Another source of errors is the mismatch between the manually segmented audio features on which the HMM parameters were estimated, and the automatically generated audio features. An improvement was obtained by estimating the HMM parameters using automatically generated audio features, but it did not outclass the video only system.

Nevertheless, integrating multiple cues from different media improves performance. In particular, ball hits detection improves classification rates for rallies and 1st serves. And music detection prevents from false detection of breaks.

5. CONCLUSION

In this paper, we have presented an audio-visual model for tennis video structure parsing. Tennis structure is modelled by an HMM that integrates a priori information about tennis content and editing rules. HMMs provide an efficient framework to merge audio and visual cues, and to perform simultaneously classification and segmentation into scenes. Experiments show that audio information improve the performance of video classification, when features are quite reliable. Indeed errors from feature extraction spread over the entire analysis process. It is therefore necessary to enhance audio segmentation and classification.

To extend our tennis video structure analysis, we also intend to include a more sophisticated audio-video fusion model by refining audio and video content analysis.

6. REFERENCES

- [1] W. Hua, M. Han, and Y. Gong, "Baseball scene classification using multimedia features," in Proc. of Int'l Conf. on Multimedia and Expo, 2002.
- [2] W. Qi, L. Gu, H. Jiang, X-R. Chen, and H-J. Zhang, "Integrating visual, audio and text analysis for news video," in Proc. of Int'l Conf. on Image Processing, 2000.
- [3] B. Li and I. Sezan, "Event detection and summarization in american football broadcast video," in SPIE Storage and Retrieval for Media Databases, v. 4676, pp. 202-213, 2002.
- [4] K. Kim, J. Choi, N. Kim, and P. Kim, "Extracting semantic information from basketball video based on audio-visual features," in Proc. of Int'l Conf. on Image and Video Retrieval, vol. 2383, pp. 278-288, 2002.
- [5] J. Assfalg, M. Bertini, A. Del Bimbo, W. Nunziati, and P. Pala, "Soccer highlights detection and recognition using hmms," in Proc. of Int'l Conf. on Multimedia and Expo, 2002.
- [6] P. Chang, M. Han, and Y. Gong, "Extract highlights from baseball game video with hidden markov models," in Proc. of Int'l Conf. on Image Processing, 2002.
- [7] E. Kijak, L. Oisel, and P. Gros, "Temporal structure analysis

of broadcast tennis video using hidden markov models,? in
SPIE Storage and Retrieval for Media Databases, v. 5021,
pp. 289?299, 2003.

11. assistant (2026-06-06T17:38:00.214Z)

I have the full primary source text. Let me verify the claim against it point by point.

****The exact quote from the paper (Page 4):****

> "The approach using only audio features automatically segmented gives poor performance. Recall rate for audio segmentation as described in 2.3 is 76.9%, and precision is 46.3%. Ball hits and music are well classi?ed, whereas noise is often classi?ed as ball hits. The errors from the automatic audio segmentation spread over the structuration process and drag the performance down... An improvement was obtained by estimating the HMM parameters using automatically generated audio features, but it did not outclass the video only system."

Now let me check each element of the claim against the source.

****Claim breakdown:****

1. "Audio-based ball-hit detection is unreliable in isolation" ? The claim's framing. The paper measures the *audio segmentation classifier* (speech/applause/ball-hits/noise/music), not ball-hit detection alone.
2. "their fully-automatic audio segmentation (speech/applause/ball-hits/noise/music classifier) achieved only 76.9% recall and 46.3% precision" ? Section 2.3 confirms exactly these 5 classes; Page 4 confirms recall 76.9%, precision 46.3%. ****Exact match.****
3. "noise frequently misclassified as ball hits" ? Paper: "noise is often classi?ed as ball hits." ****Match**** (paraphrase faithful).
4. "extraction errors propagated downstream, dragging the audio-only structuring below the visual-only system" ? Paper: "errors... spread over the structuration process and drag the performance down" and "did not outclass the video only system." Table 1 confirms: audio-segm segmentation accuracy = 44% vs visual = 71%; audio-segm rally recall = 28% vs visual rally recall = 54%. ****Match.****

Now let me verify there's no important nuance the claim omits ? specifically whether the paper actually supports a strong "audio is unreliable in isolation" conclusion, or whether audio was actually shown to be *useful*.

The critical nuance: The same paper found that ****manually-annotated**** audio features (clean labels) performed ***very well*** ? rally precision 95%/recall 63%, and the combined audio-visual system with manual audio beat everything (rally precision 96%/recall 83%). The paper's actual conclusion is ***not*** "audio is useless" but "audio feature extraction needs improvement; the model is sound, the extractor is the weak link." The paper explicitly says: "audio information improve the performance of video classi?cation, when features are quite reliable" and "ball hits detection improves classi?cation rates for rallies and ?rst serves."

Let me search for any contradicting or qualifying evidence on the broader claim about audio-based hit detection reliability, since the research question is about whether audio hit-detection is viable today.

12. user (2026-06-06T17:38:08.505Z)

Web search results for query: "racket sport ball hit detection from audio deep learning onset detection performance"

Links: [{"title": "Shot Detection in Racket Sport Video at the Frame Level Using A Recurrent Neural Network | Request PDF", "url": "https://www.researchgate.net/publication/340693456_Shot_Detection_in_Racket_Sport_Video_at_the_Frame_Level_Using_A_Recurrent_Neural_Network"}, {"title": "Spin Detection Using Racket Bounce Sounds in Table Tennis This research was funded by Sony AI.", "url": "https://arxiv.org/html/2409.11760v1"}, {"title": "A comprehensive review of ball detection techniques in sports", "url": "https://peerj.com/articles/cs-3079.pdf"}, {"title": "A comprehensive review of ball detection techniques in sports - PMC", "url": "https://pmc.ncbi.nlm.nih.gov/articles/PMC12453710/"}, {"title": "(PDF) Detection of ball hits in a tennis game using audio and visual information", "url": "https://www.researchgate.net/publication/26119682_Detection_of_ball_hits_in_a_tennis_game_using_audio_and_visual_informat"}, {"title": "Multi-Modal Hit Detection and Positional Analysis in Padel Competitions", "url": "https://www.researchgate.net/publication/340693456_Shot_Detection_in_Racket_Sport_Video_at_the_Frame_Level_Using_A_Recurrent_Neural_Network"}]

https://openaccess.thecvf.com/content/CVPR2024W/CVsports/papers/Decorte_Multi-Modal_Hit_Detection_and_Positional_Analysis_in_Padel_Competitions_CVPRW_2024_paper.pdf},{\"title\":\"Building a Robot Umpire with Deep Learning Video Analysis | The Hardball Times\",\"url\":\"https://tht.fangraphs.com/building-a-robot-umpire-with-deep-learning-video-analysis/\"},{\"title\":\"Ball Hit Detection in Table Tennis Games Based on Audio Analysis | IEEE Conference Publication | IEEE Xplore\",\"url\":\"https://ieeexplore.ieee.org/document/1699506/\"},{\"title\":\"(PDF) Ball Hit Detection in Table Tennis Games Based on Audio Analysis\",\"url\":\"https://www.researchgate.net/publication/224649455_Ball_Hit_Detection_in_Table_Tennis_Games_Based_on_Audio_Analysis\"}]

I've found relevant research on racket sport ball hit detection using audio and deep learning. Here are the key findings:

Deep Learning Approaches for Hit Detection

Deep learning-based hit detection models have been adapted from SED-net (Sound Event Detection architecture), which is particularly suited for this task since the detection and classification of racket-ball bounce sound falls in the field of Sound Event Detection (SED), which consists of detecting the temporal onset and offset of different sound events and classifying them.

Performance Metrics

Research has achieved notable performance levels:

- Using Energy Peak Detection (EPD) and Mel Frequency Cepstral Coefficient-based (MFCC-based) Refinement (MBR), high precision (91%) and adequate recall (73%) of ball hit detection are achieved
- One machine learning algorithm showed a detection accuracy higher than 95%, while the average accuracy of the whole system applied on-court was 85%

Multi-Modal Approaches

Recent work combines audio with visual information. A multi-modal algorithm for hit detection in padel utilizes audio and video extraction features, using audio to identify ball hits and perform ball detection from video feed only around the time of recognition to tackle challenges like occlusion.

Dataset Annotation

Researchers annotate the onset and offset times of hits, with studies annotating thousands of hits (2377 hits in one dataset).

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

13. user (2026-06-06T17:38:09.173Z)

Web search results for query: "audio-based shot hit detection badminton tennis accuracy precision recall sound"

Links: [{\"title\":\"(PDF) Ball Hit Detection in Table Tennis Games Based on Audio Analysis\",\"url\":\"https://www.researchgate.net/publication/224649455_Ball_Hit_Detection_in_Table_Tennis_Games_Based_on_Audio_Analysis\"},{\"title\":\"Wearable Audio and IMU Based Shot Detection in Racquet Sports | Request PDF\",\"url\":\"https://www.researchgate.net/publication/325168095_Wearable_Audio_and_IMU_Based_Shot_Detection_in_Racquet_Sports\"},{\"title\":\"Wearable Audio and IMU Based Shot Detection in Racquet Sports\",\"url\":\"https://arxiv.org/pdf/1805.05456\"},{\"title\":\"[1805.05456] Wearable Audio and IMU Based Shot Detection in Racquet Sports\",\"url\":\"https://arxiv.org/abs/1805.05456\"},{\"title\":\"(PDF) Detection of ball hits in a tennis game using audio and visual information\",\"url\":\"https://www.researchgate.net/publication/26119682_Detection_of_ball_hits_in_a_tennis_game_using_audio_and_visual_information\"},{\"title\":\"Detection of Ball Hits in a Tennis Game Using Audio and Visual Information\",\"url\":\"http://www.apsipa.org/proceedings_2012/papers/316.pdf\"},{\"title\":\"Detection of Tennis Events from Acoustic Data | Proceedings Proceedings of the 2nd International Workshop

on Multimedia Content Analysis in Sports", "url": "https://dl.acm.org/doi/10.1145/3347318.3355520"}, {"title": "Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera", "url": "https://www.mdpi.com/1424-8220/24/13/4372"}, {"title": "Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera - PMC", "url": "https://pmc.ncbi.nlm.nih.gov/articles/PMC11244353/"}]

I found several relevant research studies on audio-based shot hit detection in racquet sports. Here are the key findings:

Audio-Based Detection Systems

For table tennis games, employing energy peak detection (EPD) and Mel frequency cepstral coefficient-based refinement (MBR) achieved high precision (91%) and adequate recall (73%) of ball hit detection.

In another table tennis study, a lightweight Energy Peak Detection approach achieved an F-score of 81%.

Fusion-Based Approaches

A wearable system combining IMU and audio sensor data for shot detection achieved 95.6% accuracy. This approach was particularly valuable for table tennis and badminton, which are relatively less explored in motion sensor-based analysis due to possibly less intense hand motion during shots.

Tennis-Specific Results

A system detecting tennis events from acoustic data achieved an F1-score of 92.39% for atomic tennis events and average precision and recall values of around 80% for point boundaries.

Another tennis-based system combining audio and visual information showed a detection accuracy higher than 95% for the machine learning algorithm, though the on-court system accuracy was 85%.

These studies demonstrate that audio-based shot detection has achieved varying levels of success across different racquet sports, with accuracy generally ranging from 73% to 95%+ depending on the methodology and conditions.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

14. assistant (2026-06-06T17:38:26.922Z)

The search results are highly informative and reveal an important nuance for assessing the *strength* of the claim. Let me assess.

Verification summary:

The Kijak 2003 quote is **verbatim accurate**. I confirmed the exact text from the primary PDF:

- Section 2.3 confirms the 5 classes: "speech, applause, ball hits, noise and music"
- Page 4 confirms: "Recall rate for audio segmentation... is 76.9%, and precision is 46.3%. Ball hits and music are well classified, whereas noise is often classified as ball hits. The errors from the automatic audio segmentation spread over the structuration process and drag the performance down... An improvement was obtained by estimating the HMM parameters using automatically generated audio features, but it did not outclass the video only system."
- Table 1 confirms the downstream degradation: audio-segm segmentation accuracy 44% vs visual 71%.

So the **narrow factual claim** is fully supported by the quote and primary source. The claim is well-sourced (primary, peer-reviewed ICME conference paper).

****However****, the adversarial task is to assess whether the claim, ***as deployed in the research context***, is misleading or overreaching. Two problems:

1. ****The claim's headline framing ("Audio-based ball-hit detection is unreliable in isolation") is an overgeneralization the source does not support.**** Critically, ***in the very same paper***, the audio classifier said *****"Ball hits and music are well classified"***** ? i.e., ball-hit detection itself was NOT the weak link. The poor 46.3% precision was driven by ****noise being misclassified as ball hits**** (a class-confusion / false-alarm problem on a 5-way classifier), not by ball-hit detection being inherently unreliable. And the manually-annotated audio features performed strongly (rally precision 95%). The paper's own conclusion is that audio ***improves*** performance "when features are quite reliable" and "ball hits detection improves classification rates for rallies." So using this paper to argue against audio-based hit detection inverts the paper's actual finding.

2. ****The claim is outdated for the field's current state (2003 vs 2026).**** Modern audio hit-detection work directly contradicts the implied "audio is too unreliable" conclusion: table-tennis audio EPD+MBR reaches ****91% precision / 73% recall****; tennis acoustic event detection reaches ****F1 92.4%****; SED-net deep models and multimodal (audio+video) padel/tennis systems reach 85-95%+. The 2003 result used Gaussian-mixture Viterbi segmentation, a method obsolete by modern SED standards.

This is voter 3/3 with an adversarial mandate. The narrow quoted fact is accurate, but the claim ****as a research conclusion**** ("audio-based ball-hit detection is unreliable in isolation") is a misread/overreach of the source ? the paper actually says ball hits ***were*** well classified and audio ***helps***, and the result is 23 years old in a fast-moving field with strongly contradicting modern evidence. The strong generalizing headline is not supported by the quote, which is the first checklist failure, compounded by the outdated-for-field-state failure.

15. user (2026-06-06T17:38:41.302Z)

Structured output provided successfully

16. assistant (2026-06-06T17:38:49.553Z)

Verification complete. The narrow quoted numbers (76.9% recall, 46.3% precision, noise?ball-hit confusion) are verbatim-accurate against the primary PDF I extracted, but I refuted the claim as a research conclusion: it overreaches the source (the paper says ball hits were "well classified" and audio ***improves*** results when features are reliable ? the failure was the 2003 GMM/Viterbi extractor and noise false-alarms, not ball-hit detectability), and it's 23 years outdated for a fast-moving field where modern audio hit-detection reaches 91% precision / F1 92%+.